

BUILT IN FUNCTIONS

Python has the following built in functions.

1) print() function:

It is used to display output messages on monitor.

Examples:

1) `print("Welcome")`

2) `a=10`

`print(a)`

3) `a,b,c=10,20,30`

`print(a,b,c)`

4) `a,b,c=10,20,30`

`print(a,b,c, sep=", ")`

5) `a=b=c=10`

`print(a, b, c, sep=", " end=".")`

6) `a=10;b=20;c=30`

`print(a, b, c)`

7) `a=10`

`b=20`

`print(a)`

`print(b)`

2) type() function:

It is used to get the type of an object.

Examples:

1) `a=10`

`type(a) (or) print(type(a))`

2) `a=True`

`type(a) (or) print(type(a))`

3) `a="Hello"`

`type(a) (or) print(type(a))`

4) `a=10.2`

`type(a) (or) print(type(a))`

```
5) a=3+2j  
   type(a) (or) print(type(a))
```

3) int() function:

It returns an integer number.

4) float() function:

It returns a floating point number.

5) complex() function:

It returns a complex number.

6) str() function:

It returns a string.

7) bool() function:

It returns a boolean value.

The above int(), float(), complex(), str() & bool() functions are called as type conversion functions.

Note: Complex numbers cannot be converted to integers & floating point numbers.

8) input() function:

It is used to get the data from keyboard. It returns str format.

Examples:

```
1) a=input("Enter first number: ")  
   b=input("Enter second number: ")  
   c=int(a)  
   d=int(b)  
   e=a+b  
   print(e)
```

- 2)

```
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
c=a+b
print(c)
```
- 3)

```
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
print(a+b)
```
- 4)

```
print(int(input("Enter first number: "))+
      int(input("Enter second number: ")))
```

9) id() function:

It is used to get the id of an object.

Examples:

- 1)

```
a=10
id(a) (or) print(id(a))
```
- 2)

```
a="Hello"
id(a) (or) print(id(a))
```

10) bin() function:

It is used to get the binary value of a given number.

Examples:

- 1)

```
a=10
b=bin(a)
print(b)
```

Binary literals are prefixed with 0b (or) OB

Examples:

- 1)

```
a=0b1000
print(a)
```

11) oct() function:

It is used to get the octal value of a given number.

Examples:

```
1) a=40
   b=oct(a)
   print(b)
```

Octal literals are prefixed with 0o (or) 0O

Examples:

```
1) a=0o36
   print(a)
```

12) hex() function:

It is used to get the hexadecimal value of a given number.

Examples:

```
1) a=40
   b=hex(a)
   print(b)
```

Hexadecimal literals are prefixed with 0x (or) 0X

Examples:

```
1) a=0x1e
   print(a)
```

13) chr() function:

It is used to get the character of a given ascii number.

Examples:

```
1) a=65
   b=chr(a)
   print(b)
```

14) ord() function:

It is used to get the ascii value of a given character.

Examples:

```
1) a='A'  
   b=ord(a)  
   print(b)
```

Python programs can be written in two ways:

- 1) Interactive Mode
- 2) Batch Mode

Interactive Mode:

The concept of submitting statement by statement to python interpreter is called as interactive mode.

In order to write programs in interactive mode we require python shell or python IDLE(Integrated Development Learning Environment).

Interactive mode is not suitable for developing applications & projects.

It is suitable for learning python. It is also suitable to execute code snippets.

Batch Mode:

The concept of submitting all statements at a time to python interpreter is called as batch mode.

In order to write programs in batch mode we require editors, notepad, notepad++, edit plus & IDEs(Integrated Development Environment).

It is suitable for learning & developing applications and projects also.

By

Mr. Venatesh Mansani